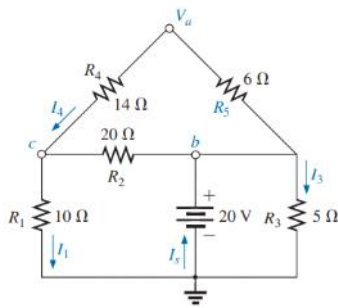
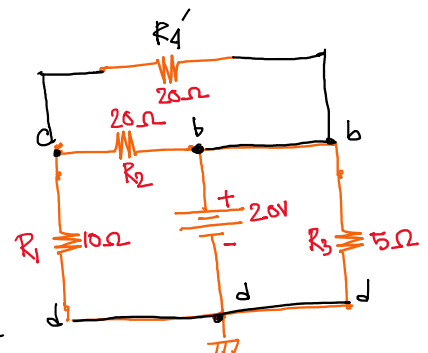
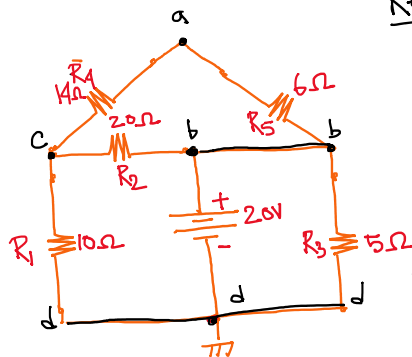




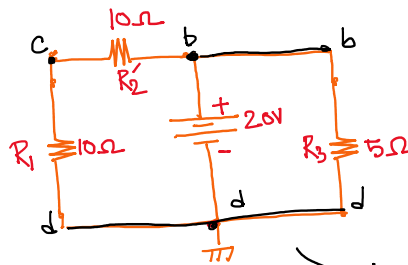
Reduce

R_4 & R_5 are in series
 $R_4' = R_4 + R_5 = 14 + 6 = 20\Omega$

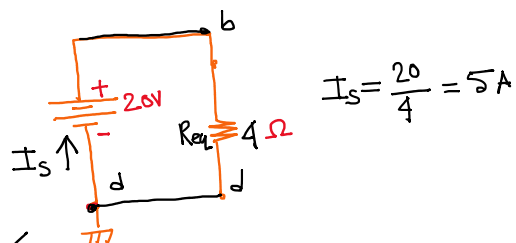
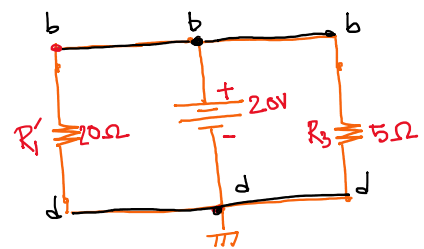
R_2 & R_4' are parallel.
 $R_2' = R_2 \parallel R_4' = 20\Omega \parallel 20\Omega = 10\Omega$



$R_1' = R_1 + R_2' = 10 + 10 = 20\Omega$

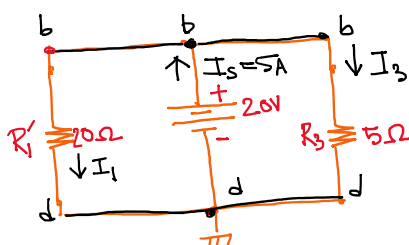


$R_{eq} = R_1' \parallel R_3 = 20 \parallel 5 = 4\Omega$



$$I_s = \frac{20}{4} = 5A$$

Return

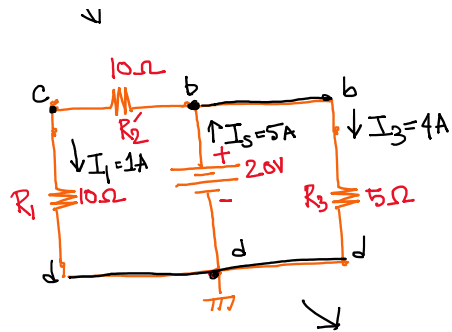


$$I_3 = \frac{R_1'}{R_1 + R_3} I_s \quad [\text{Current divider Rule}]$$

$$= \frac{20}{25} \times 5 = 4A$$

$$I_1 = \frac{R_3}{R_1 + R_3} I_s = \frac{5}{25} \times 5 = 1A$$

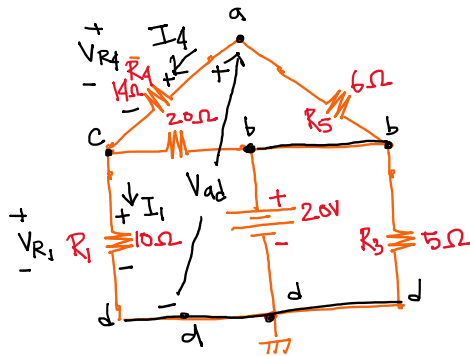
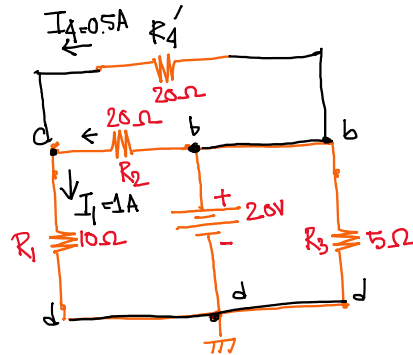




$$I_4 = \frac{R_2}{R_2 + R_4} I_1$$

$$= \frac{20}{20 + 20} 1$$

$$= 0.5A$$



To determine V_a :

$$V_a = V_{ad} = V_a - V_d$$

$$= V_a$$

V_d : grounded
 $V_d = 0V$

Apply KVL in the loop $a-b-d-a$:

$$+V_{R1} + V_{R4} - V_{ad} = 0 \Rightarrow I_1 R_1 + I_4 R_4 = V_{ad} = V_a \quad [V_d = 0V]$$

$$\Rightarrow V_a = 10 \times 1 + 14 \times 0.5 = 10 + 7 = 17V$$